

AESHA: Accelerating Eigen-decomposition-based Sparse Transformer with Hybrid RRAM-SRAM Architecture

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Abstract

Compute-in-memory (CIM) architectures based on emerging non-volatile memories (eNVM) are recognized as promising candidates for the efficient computation of self-attention-based Transformer models, which are bounded by memory-intensive dynamic computations involving large matrices. However, existing CIM-based approaches mostly focused on the acceleration of vector-matrix multiplications (VMM) to obtain Q , K and V matrices for direct or sparse calculations of the vanilla Transformer. By storing large amounts of intermediate results in eNVM, the endurance limit is ignored which contradicts the reality of these technologies. In this work, a hybrid RRAM-SRAM CIM architecture to accelerate Transformers, namely AESHA, is proposed based on the eigen-decomposition of $W_Q W_K^T$ and the systolic in-memory attention reconstruction. By utilizing the features of symmetric/skew-symmetric real matrices, AESHA translates dynamic attention computation to RRAM-based sparse feature transformation and SRAM-based systolic attention reconstruction to fully exploit the advantages of both on the architectural level. Experiments on a broad spectrum of benchmarks demonstrate that AESHA delivers superior performance in terms of pipeline optimization and energy efficiency. Specifically, AESHA achieves 2.59×, 2.69×, 2.62×, 2.58×, 3.08× and 7.84× maximum static RRAM memory footprint reduction in BERT-Base, BERT-Large, BigBird, Sanger, ViT-Base and ViT-Large over vanilla computation stack, respectively. It eliminates all runtime write access to the RRAM arrays to circumvent the endurance limit. Additionally, AESHA demonstrates 3170.0×, 95.6×, 4.1×, 19.0×, and 19.2× speedup, and 336.0K×, 12.6K×, 7.3×, 23.1×, and 25.8× energy reduction over CPU, GPU, ReBERT, ReTransformer, and CPSAA, respectively.

Keywords

Eigen Decomposition, Self-attention Acceleration, RRAM-SRAM Hybrid Architecture, Compute-in-memory

1 Introduction

The rapid advancement of artificial intelligence (AI) is fundamentally transforming the electronics industry, with technologies like AI-powered content creation, autonomous vehicles and smart home increasingly reshaping our daily lives. As a pioneer in self-attention

neural networks, Transformer has demonstrated sweeping success across natural language processing [1, 2], computer vision [3] and multi-modal cognitive tasks [4]. Notably, Transformers adeptly capture long-range correlations of tokenized inputs (e.g., sentences or images) through the self-attention mechanism, thereby resulting in a global receptive field and enhanced learning capability. However, self-attention's computational complexity and memory requirements grow quadratically with input dimension, making it less efficient on conventional computational hardware.

In order to improve the energy efficiency and inference speed of Transformers on resource-constrained devices, extensive endeavors in dedicated accelerators based on FPGA or ASIC have been reported which covered a wide range of design options and optimization techniques. One endeavor prioritizes reducing computational overhead and memory footprint through layer-level, head-wise, or token-grained entry pruning based on softmax importance or entropy criteria [5–8]. At the same time, another strategy employs low-rank linear approximations or low-precision network predictions for weak relationship detection at runtime, thereby omitting unnecessary computations [9–12]. A further strategy involves reordering the sparse attention matrix through packing, splitting, grouping, or blocking to regularize the attention matrix and enable its subsequent efficient computations [13, 14]. Despite their exceptional performances, these works still fall into the category of von-Neumann architectures featuring separated processing and memory units (typically DRAM). These architectures necessitate continuous retrieval of embeddings and weights from off-chip DRAM, creating a bottleneck for on-chip query-key computations.

Recently, CIM architectures have emerged as a new research direction to accelerate Transformer computations with higher energy efficiency. These architectures merge the computation and model storage into the same CIM array, effectively reducing off-chip memory access and associated energy consumption. A substantial body of accelerators employ conventional memory implementations (such as SRAM and Flash) as well as eNVMs like RRAM, MRAM, PCM and FeRAM to accelerate VMM. Some studies [15–18] explore eNVM-friendly data access through techniques like matrix decomposition or fine-grained dataflow designs to accelerate the VMM primitive in attention computation. Other studies [15, 19] focus on optimizing the maximum value search in softmax by investigating logic-in-memory or content-addressable matching. In addition to these approaches, some research [20–23] employs heterogeneous RRAM-SRAM CIM designs to balance low-precision prediction and exact computation workloads towards providing low-power and efficient solutions.

Despite the potential of CIM-based designs for Transformers, it is often difficult to strike a fair balance between accelerating the attention computation and minimizing the memory footprint. Heavy

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stationary memory footprint and frequent intermediate memory write pressure persist in most prior arts. Some approaches [24, 25] reduce on-chip memory occupancy of partial feature vectors by confining attention windows or blocks. However, this comes at the sacrifice of the model's accuracy. Alternative methods such as SPRINT [20] and HARDSEA [21] leverage dynamic sparse attention features to reduce ineffective computations. Nevertheless, these methods still necessitate preserving entire feature vectors for attention recomputation. In this context, exploring a structured prunable computation pipeline and acceleration architecture remains crucial.

Instead of solely optimizing the pipeline based on the matrix transposition property and associative law of matrix multiplication, we leverage the quadratic coupling of input concerning attention weights in self-attention to design a novel pipeline named AESHA. AESHA aims to reduce static weight and intermediate activations occupancy through structured pruning. Furthermore, we propose a heterogeneous CIM-based architecture to accelerate attention reconstruction by combining the merits of RRAM and SRAM. The major contributions of this work are summarized as follows:

- We propose a novel eigen-decomposition-based attention computation pipeline, which equivalently transforms attention computation into the orthogonal feature transformation plus the outer-product-based attention reconstruction to optimize the dataflow's compatibility with CIM.
- We propose an adaptive energy-distribution threshold-based pruning strategy, which achieves feature space compression with principal feature vector retention in head granularity, thus effectively reducing the memory occupation of static weights and intermediate activations.
- We introduce a heterogeneous CIM implementation featuring a hybrid dataflow for versatile computation tasks in the eigen-decomposition-based attention paradigm. Specifically, we utilize analog weight-stationary (WS) based RRAM-CIM for feature transformations, and digital output-stationary (OS) based SRAM-CIM for attention reconstruction.
- Extensive experiments and ablation studies on various transformer models consistently validate the advantages of our proposed AESHA framework, with throughput and energy efficiency advantages over both general computing platforms CPUs, GPUs, and prior RRAM-based Transformer accelerators ReBERT [24], ReTransformer [15] and CPSAA [16].

2 Background and Motivations

2.1 Background

Self-Attention in Transformers. Fig. 1(a) illustrates the transformer architecture consisting of multi-head self-attention (MHSA) and feed-forward network (FFN) modules. The flow of vanilla self-attention is further elucidated in Fig. 1(b)-(i). First, three feature matrices from the same token sequence are calculated via linear transformation, namely query (Q), key (K), and value (V). Each feature is a dense matrix with size $\mathbb{R}^{N \times d}$, where N is the token number and d is the head-wise feature size. Next, the attention score matrix $S \in \mathbb{R}^{N \times N}$ is computed by taking the dot-product of query and key, the element of which indicates the similarity between a specific pair of tokens. Subsequently, the row-wise Softmax normalizes the attention scores and enhances the largest elements. In accelerator

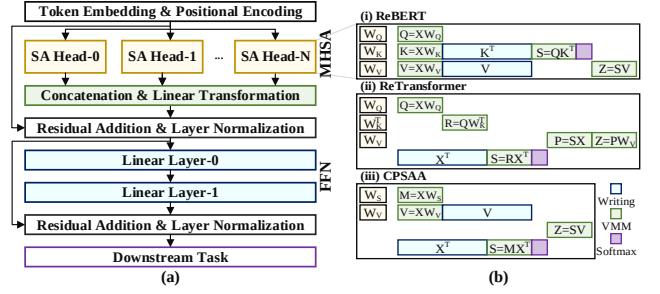


Figure 1: (a) Transformer structure. (b) Attention computation pipeline, i-iii denote vanilla self-attention (SA), ReTransformer [15], and CPSAA [16] respectively.

designs, masked Softmax (see Equation 1) are often used to skip weak-related tokens for more efficient attention aggregation.

$$\hat{S}_{ij} = \frac{\exp(S_{ij}) \odot M_{ij}}{\sum_{j=1}^N \exp(S_{ij}) \odot M_{ij}}, M_{ij} = \begin{cases} 1 & \text{if } S_{ij} \geq \theta \\ 0 & \text{if } S_{ij} < \theta \end{cases} \quad (1)$$

where θ denotes the threshold values and the mask M_{ij} redistributes the importance of each token: a token contributes to context aggregation when $M_{ij} = 1$; otherwise, its influence is disregarded.

Furthermore, the output matrix $Z \in \mathbb{R}^{N \times d}$ of each head is obtained by the weighted sum of the value set proportional to measured correlations. Finally, the ensembled features of h heads are fused by a projection matrix W_O , as shown in Equation 2.

$$O = \text{Concat}(Z_1, Z_2, \dots, Z_h)W_O + b_O \quad (2)$$

The output from the MHSA module will go through the FFN, comprising successive linear layers followed by non-linear activation, which can be succinctly formulated as:

$$\text{FFN}(O) = \sigma(OW_{\text{FFN1}} + b_{\text{FFN1}})W_{\text{FFN2}} + b_{\text{FFN2}} \quad (3)$$

where W_{FFN1} , W_{FFN2} , b_{FFN1} and b_{FFN2} denote the linear weights and biases, while $\sigma(\cdot)$ is the nonlinear activation such as ReLU/GeLU.

2.2 Motivation

Fig. 1(b) outlines some notable pipeline optimization efforts in existing RRAM-based attention accelerators (ReTransformer and CPSAA). These efforts primarily focus on mitigating data compute-write-compute (CWC) dependencies or shortening the computation sequence. Nevertheless, they still contain the dimension coupling between feature weight matrices (e.g., W_Q and W_K), making it difficult to apply canonical pruning strategies of deep neural networks to compress attention weights. Even if compression scenarios are achieved by matching the dimensions of query and key weights, it is still challenging to establish optimization objectives for adaptive attention recovery because the associated low-rank approximation in attention maps may significantly lower the accuracy.

Furthermore, existing RRAM-based attention computation stacks overlook the limitation of its endurance. In these studies, the dynamic input embeddings or intermediate activation results are frequently written into the RRAM arrays, which are used as buffer memory. However, the endurance of commercial RRAM technologies is capped at approximately 10^6 cycles according to multiple sources [26, 27], which contradicts with the high writing demands of buffer memory. Minimizing the writing of intermediate results

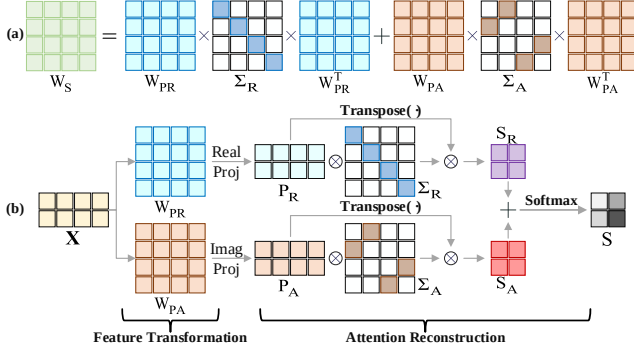


Figure 2: (a) The attention weight matrix decomposition principle and (b) AESHA pipeline.

into RRAM (or other eNVM) is imperative for improving the long-term viability of eNVM-based CIM for edge inference applications.

Thus, the primary motivation of this work is to explore a computation pipeline optimization scheme that combines enhancing computation efficiency with minimizing memory footprint. As described in the following sections, an algorithm-architecture co-optimization strategy is adopted to overcome the discrepancy between pipeline efficiency, model accuracy and endurance limitation.

3 Algorithm Optimization

3.1 Optimized Pipeline

Unlike previous pipeline optimizations, this work considers the quadratic relationship between input and attention weight matrix, introducing an eigen-decomposition-based stack to reduce computation path latency and static memory occupancy (see Fig. 2).

It is observed that the attention weight matrix $W_S \in \mathbb{R}^{D \times D}$, formed by the product of the query and key weight matrices, exhibits strong low-rank characteristics ($d \ll D$), as expressed in Equations 4-5. The low-rank feature of W_S suggests that the attention weight matrix contains redundant information, allowing reconstruction of the eigenspace with only d eigenvectors.

$$\begin{aligned} S &= Q \cdot K^T = (X \cdot W_Q) \cdot (X \cdot W_K)^T \\ &= (X \cdot W_Q) \cdot (W_K^T \cdot X^T) = X \cdot \underbrace{(W_Q \cdot W_K^T)}_{W_S} \cdot X^T \end{aligned} \quad (4)$$

$$\text{Rank}(W_S) \leq \min\{\text{Rank}(W_Q), \text{Rank}(W_K)\} \leq d \quad (5)$$

Drawing inspiration from factorization theory [28], arbitrary square matrix $W_S \in \mathbb{R}^{D \times D}$ can be uniquely decomposed into symmetric and skew-symmetric components, as expressed below:

$$W_S = \underbrace{(W_S + W_S^T)/2}_{W_R: \text{real-symmetric matrix}} + \underbrace{(W_S - W_S^T)/2}_{W_A: \text{skew-symmetric matrix}} \quad (6)$$

Expanding upon spectral-decomposition principles [29, 30], we can further decompose the two symmetric matrices using $U\Sigma U^T$ to ascertain the eigenbases of the attention weight matrix in both the real and imaginary spaces. In this context, U represents the orthogonal matrix that satisfies the property $UU^T = I$. The eigenvalues $\sigma_{R,i}$ in the real space are distributed along the major diagonal, whereas the eigenvalues $\sigma_{A,i}$ in the imaginary space are paired in 2×2

blocks symmetrically along the main diagonal, as outlined below:

$$\begin{aligned} S &= S_R + S_A = X W_S X^T = X (W_R + W_A) X^T \\ &= X (W_{PR} \Sigma_R W_{PR}^T) X^T + X (W_{PA} \Sigma_A W_{PA}^T) X^T \end{aligned} \quad (7)$$

$$\begin{aligned} &= \underbrace{(X W_{PR})}_{P_R} \Sigma_R \underbrace{(W_{PR}^T X^T)}_{P_R^T} + \underbrace{(X W_{PA})}_{P_A} \Sigma_A \underbrace{(W_{PA}^T X^T)}_{P_A^T} \\ \Sigma_R &= \text{diag}\{\sigma_{R,1}, \sigma_{R,2}, \dots, \sigma_{R,D_R}\}; D_R = 2 \times D_A \\ \Sigma_A &= \text{diag}\left\{\begin{pmatrix} 0 & \sigma_{A,1} \\ -\sigma_{A,1} & 0 \end{pmatrix}, \begin{pmatrix} 0 & \sigma_{A,2} \\ -\sigma_{A,2} & 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 & \sigma_{A,D_A} \\ -\sigma_{A,D_A} & 0 \end{pmatrix}\right\} \end{aligned} \quad (8)$$

where P_R and P_A denote the mappings of the input embeddings in the real and imaginary feature spaces, respectively. Here, D_A signifies the number of eigenvalue pairs of the skew-symmetric matrix, which is half of the embedding dimension D_R . S_R and S_A refer to the symmetric and skew-symmetric components of the attention score matrix (For more details, see Fig. 2(b)).

Capitalizing on the property that a symmetric matrix can be effectively reconstructed from the sum of the outer products of its eigenvectors, we can further translate Equation 7 into the following formulation:

$$\begin{aligned} S &= \sum_{i=1}^{D_R} \sigma_{R,i} P_R[:, i] \cdot P_R[:, i]^T + \sum_{i=1}^{D_A} \sigma_{A,2i} \{P_A[:, 2i-1] \cdot P_A[:, 2i]^T \\ &\quad - P_A[:, 2i] \cdot P_A[:, 2i-1]^T\} \end{aligned} \quad (9)$$

The above formulations effectively diagonalize the W_S matrices and lay the foundation for subsequent new architectural designs.

3.2 Energy-distribution-based Pruning Strategy

Given the low-rank structures within the attention weight matrix W_R and W_A , we propose an energy-aware pruning strategy to reduce memory footprint through head-level threshold searching.

Since the eigenvectors can be arranged to match the diagonal eigenvalues within Σ , we sort the eigenvalues by absolute magnitude in descending order, yielding the sorted eigen-matrices (i.e., W_{PR}' and W_{PA}') along with the diagonal matrices (i.e., Σ_R' and Σ_A'). The element in Σ' meets the following criteria:

$$|\sigma'_{R,1}| \geq |\sigma'_{R,2}| \geq \dots \geq |\sigma'_{R,D_R}|; |\sigma'_{A,1}| \geq |\sigma'_{A,2}| \geq \dots \geq |\sigma'_{A,D_A}| \quad (10)$$

To standardize the variability in eigenvalues, we develop an energy-separability-based criterion, namely ρ_k , as defined in Equation 11, which entails dividing the square of each eigenvalue by the sum of squares of all elements within the entire vector.

$$\rho_k = \frac{\sigma_k'^2}{\|\sigma'\|_2^2} = \frac{\sigma_k'^2}{\sum_{i=1}^N (\sigma_i')^2} \quad (\sigma_k' = \sigma'_{R,k} \text{ or } \sigma'_{A,k}) \quad (11)$$

Upon this basis, we can transform the low-rank matrix estimation into the search for energy-distribution-based threshold, denoted by $\tau \in [0, 1]$. By retaining the Top-k eigenvalues and their corresponding eigenvectors that meet the energy threshold, the symmetric matrix can be approximated as follows:

$$\begin{aligned} W_R &\approx W_{PR}^\# \Sigma_R^\# W_{PR}^{\#T}, W_{PR}^\# \in \mathbb{R}^{D \times k}, \text{diag}(\Sigma_R^\#) \in \mathbb{R}^k \\ W_A &\approx W_{PA}^\# \Sigma_A^\# W_{PA}^{\#T}, W_{PA}^\# \in \mathbb{R}^{D \times k}, \text{diag}(\Sigma_A^\#) \in \mathbb{R}^k \end{aligned} \quad (12)$$

$$k = \min\{\min\{k_1 | \sum_{i=1}^{k_1} \rho_{R,k} \geq \tau\}, \min\{k_2 | \sum_{i=1}^{k_2} \rho_{A,k} \geq \tau\}\} \quad (13)$$

where k denotes the minimum rank satisfying the threshold constraint for each pair of symmetric and skew-symmetric matrices, and $\#$ is used to indicate the pruned matrix.

Furthermore, we employ a two-stage threshold search strategy to determine the eigenspace pruning thresholds in each encoder block precisely. Initially, we set a global threshold g_τ to one for all eigen-matrices and gradually decrement it by a unit step τ_{unit} until the accuracy loss exceeds the tolerance range ϵ . Next, we initialize the thresholds for all eigen-matrices to the optimal global threshold and adopt a greedy search to determine the lower bounds of the thresholds for each attention head within each encoder block by iteratively decreasing them by τ_{unit} , aiming to maximize the pruning ratio with minimum model performance degradation. Finally, after truncating the last $N - k$ eigenbasis from the eigenspace using the fine-grained thresholds $\{\tau\}$, we can update the attention model through fine-tuning instead of training from scratch. This enables the model to automatically maximize sparsity in each attention head's eigenspace during fine-tuning.

During the fine-tuning process, W'_{PR} and W'_{PA} in each attention head are treated as free trainable parameters, hence the orthogonality of the eigenvectors will be impeded. To ensure the validity of eigen-decomposition and facilitate effective rank reduction via eigenvalue pruning, we introduce an orthogonality regularizer for W'_{PR} and W'_{PA} , as illustrated below:

$$\mathcal{L}_O(W'_{PR}, W'_{PA}) = \frac{1}{k^2} (\|W'_{PR} W'_{PR}^T - I\|_F^2 + \|W'_{PA} W'_{PA}^T - I\|_F^2) \quad (14)$$

where $\|\cdot\|_F$ and I denote the Frobenius norm and Identity matrix.

For the choice of sparsity-inducing regularizer, we employ the Hoyer regularization constraint [31] on eigenvectors $\text{diag}(\Sigma'_R)$ and $\text{diag}(\Sigma'_A)$, as depicted in Equation 15. Due to its scale-invariance, this regularizer will promote the sparsity while retaining most of the energy in the eigenspace.

$$\mathcal{L}_H(\sigma) = \frac{\|\sigma'\|_1}{\|\sigma'\|_2} = \frac{\sum_i |\sigma'_i|}{\sqrt{\sum_i \sigma_i'^2}} \quad (15)$$

In addition to the above training objectives, it's crucial to maintain the proximity between the raw symmetric matrices and the reconstructed counterparts, derived from the truncated eigen-matrices and diagonal eigenvalue vectors. Therefore, we incorporate the mean square error (MSE) loss, denoted as $\mathcal{L}_T$, into the overall objective function $\mathcal{L}_{total}$ for fine-tuning, as follows:

$$\mathcal{L}_T(W'_{PR}, \Sigma'_R, W'_{PA}, \Sigma'_A) = \|W'_{PR} \Sigma'_R W'_{PR}^T - W'_{PR} \Sigma'_R W'_{PR}^T\|_F^2 + \|W'_{PA} \Sigma'_A W'_{PA}^T - W'_{PA} \Sigma'_A W'_{PA}^T\|_F^2 \quad (16)$$

$$\mathcal{L}_{total} = \mathcal{L}_T + \lambda_O \mathcal{L}_O + \lambda_H \mathcal{L}_H \quad (17)$$

where λ_O and λ_H represent the balance parameters for the orthogonality loss and sparsity-inducing regularization loss respectively.

By this point, we can implement the lightweight model deployment. Initially, we parallelize input feature transformation onto three distinct feature spaces, namely W_{PR} , W_{PA} , and W_V . Subsequently, the outer product accumulation of input-related features is achieved. Finally, attention vectors are fetched for context aggregation. Through the co-optimization of algorithms and architecture,

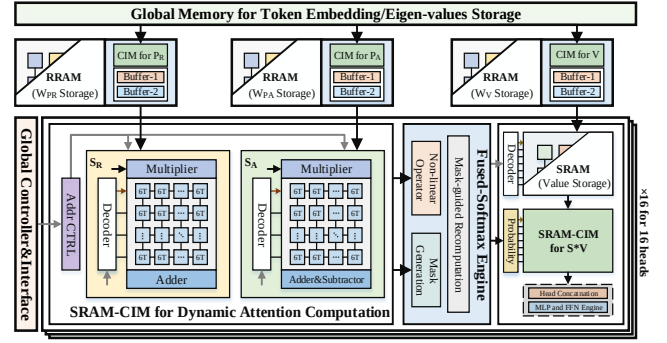


Figure 3: AESHA accelerator architecture.

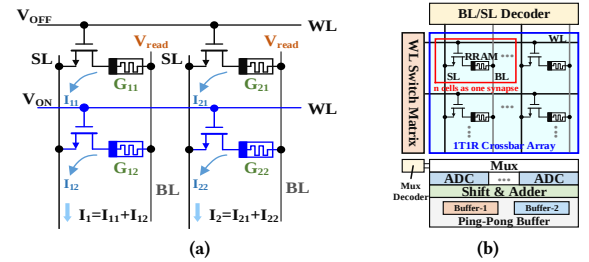


Figure 4: (a) The principles of multiply-and-accumulate (MAC) being performed along a bit line and (b) high-level schematic diagram of the synaptic array.

this new computing paradigm effectively shortens the attention computation path and reduces the memory writing burden.

4 Hardware Accelerator

4.1 Overall Architecture

The overall architecture of AESHA (see Fig. 3) is designed with sixteen cores that enable concurrent processing of self-attention heads for feature transformation and attention reconstruction. For the feature transformation phase, the compressed feature weight matrices (e.g., W'_{PR} , W'_{PA} and W_V) are characterized by constant and tiny memory footprint, making it advantageous for efficient mapping and VMM acceleration using RRAM-CIM. On the contrary, the feature embeddings in attention computation demand frequent memory write accesses, which can degrade RRAM-CIM durability and computational efficiency. Therefore, we employ the WS-based RRAM-CIM to implement feature transformations and OS-based SRAM-CIM to accelerate dynamic attention reconstruction.

Following the completion of attention reconstruction, the results are retrieved row-by-row to facilitate the attention probability transformation in the fused-softmax engine. These mapped attention probabilities then act as attention weights for context aggregation within the WS-based SRAM-CIM macro.

4.2 RRAM-based Feature Transformation

The RRAM-CIM employs a WS workflow to minimize off-chip weight accesses and enable in-situ computation by programming the compressed attention weight matrices (e.g., W'_{PR} and W'_{PA}) and the value matrix W_V into the synaptic arrays as conductance using linear interpolation (as depicted in Equation 18).

$$g = g_{\min} + \frac{(w - w_{\min})(g_{\max} - g_{\min})}{w_{\max} - w_{\min}} \quad (18)$$

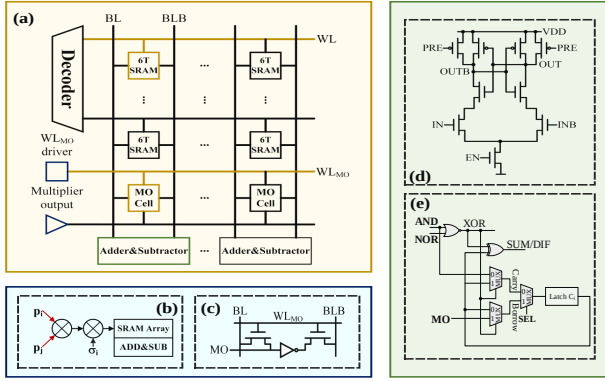


Figure 5: Attention reconstruction engine. (a) SRAM-based in-situ accumulation circuit (b) Basic processing element (c) Multiplier output cell (d) SA-based LPU (e) Co-optimized full adder-subtractor.

where g represents the mapped value of weight w in the conductance domain, $g_{\min}$ and $g_{\max}$ respectively denote the minimum and maximum conductance values, while $w_{\min}$ and $w_{\max}$ represent the minimum and maximum values of the weight.

Subsequently, the token embeddings are converted into a binary-series analog voltage (i.e., V_{ON} or V_{OFF}) and fed to the WLs, while a read voltage (V_{read}) is applied to the BLs. When the input value is set to one, the MOS transistor is turned on, causing a voltage drop across the RRAM and current flow from drain to source. Otherwise, the transistor is turned off with no current flow. The currents from multiple synaptic cells accumulate on the same shared SL, representing the dot product of the input and weight vectors (see Fig. 4(a)). Analog results generated on the BLs are converted using a 5-bit ADC. To save power and area footprint, every eight SLs share the same 5-bit ADC through time-division multiplexing, as shown in Fig. 4(b). It is worth mentioning that this binary input scenario and RRAM-CIM architecture simplify the input circuit design and eliminate sneak-path currents, while AESHA is also compatible with other RRAM-CIM architectures with multi-bit inputs.

The resulting real and imaginary attention components (P_R and P_A) will be transferred to the in-SRAM outer-product-based attention reconstruction module via ping-pong buffers. At the same time, the value set V is written to conventional SRAM synaptic arrays for subsequent context aggregation in a parallel manner.

4.3 SRAM-based Attention Reconstruction

To support outer-product-accumulation primitive in AESHA, we design an SRAM-based attention reconstruction engine (see Fig. 5(a)) specifically tailored to the attention score dataflow, which effectively circumvents pipeline stalls and eliminates power consumption associated with writing K^T or X^T to RRAM. This engine employs two separate cores to reconstruct the symmetric and skew-symmetric components of the attention score matrix in parallel. Here we use the computation of the symmetric attention matrix to elucidate how dynamic attention reconstruction works.

Fundamental Working Principles. The basic processing element (PE) comprises a two-level digital multiplier and an in-SRAM accumulator. Initially, the feature vectors of each token enter the digital multiplier within PE in a skewed fashion under clock control to perform the element-wise dot-product computation of $p_{m,i} \cdot p_{n,i}$

Table 1: SA-based computation primitive

MODE	IN	INB	OUTB	OUT
0	—	—	—	—
1	BL	$\overline{BL}$	$\overline{OUT}$	OUT
2	BL	V_{ref}	$\overline{AB}$	AB
3	V_{ref}	$\overline{BL}$	$A + B$	$A + B$

and $\sigma_i \cdot (p_{m,i} \cdot p_{n,i})$, as illustrated in Fig. 5 (b). Next, specific word-lines of SRAM and multiplication output (MO) cells are gated for bit-wise value comparison between these activation units. Subsequently, the sensing amplifier (SA) based logic processing unit (LPU) detects voltage transitions on bit-lines (BLs) to perform in-situ addition or subtraction. Finally, the accumulation results are restored in SRAM cells to update the attention scores. The reconstruction of the real and imaginary attention components finishes when all feature embeddings pass through the engine.

Since skew-symmetric attention component reconstruction involves subtraction, we implement in-situ addition and subtraction primitive by introducing control logic on an SA-based full adder [32]. The process mainly consists of MO&SA-based logic comparison and full addition and subtraction arithmetic.

● **MO&SA-based logic comparison** (see Fig. 5(c-d)). Bit-wise comparison of digital dot-product results with SRAM cell scores is done by activating WL and WL_{MO}. BL's voltage drop is constant when both cells are zero-value, and vice versa. The BL voltage ranges from 0 to Vdd when the two ends of MOS transistor have opposing values, causing discharge. After completing the comparison stage, SA-based LPUs detect voltage changes on BLs and perform bitwise AND/OR operations by enabling EN (see Fig. 5(d)). The SA-based circuitry supports ordinary read mode and two logical primitives (see Tab. 1).

● **Full addition and subtraction arithmetic** (see Fig. 5(e)).

We implement the logic primitives for a 1-bit complete adder and subtractor according to Equation 19:

$$\begin{aligned} \text{SUM} &= A \oplus B \oplus C, \text{ Carry} = AB + (A \oplus B)C; \\ \text{DIF} &= A \oplus B \oplus C, \text{ Borrow} = \overline{AB} + \overline{A \oplus B}C \end{aligned} \quad (19)$$

However, integrating the logic AND operation for inputs A and B into the SRAM read circuitry remains difficult. As stated in Equation 20, we simplify the carry and borrow output logical expression by employing $A \oplus B$ as a judgement condition to mitigate this risk.

$$\begin{aligned} \text{Carry} &= C \text{ if } A \oplus B = 1 \text{ else } AB; \\ \text{Borrow} &= B \text{ if } A \oplus B = 1 \text{ else } C \end{aligned} \quad (20)$$

where A and B represent a pair of 1-bit operands, C denotes the carry-in and borrow-in bit. SUM and DIF denote the results of addition and subtraction. As shown in Fig. 5(e), the carry-out and borrow-out functions of the 1-bit full adder-subtractor is transformed into an XOR-based binary multiplexer. When the SEL is set to one, the subtractor is enabled. The latch is employed to retain carry-in and borrow-in with initialized values of zero.

Fine-grained Array for Attention Reconstruction. We use the designed PE to construct the attention score stationary-based systolic array of size $N \times N$ for outer-product accumulation primitive in the AESHA pipeline (see Fig. 6(a)). However, there are insufficient on-chip systolic cells to implement the entire attention reconstruction in the systolic array for more challenging datasets

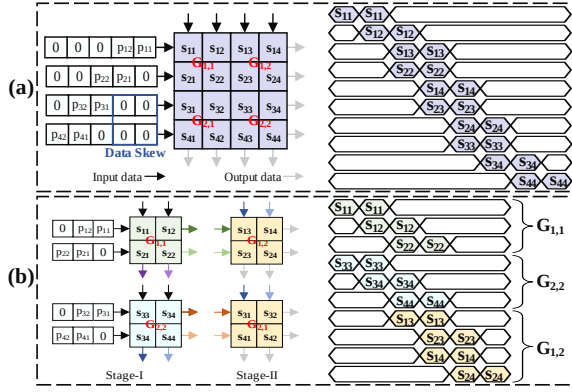


Figure 6: (a) Coarse-grained systolic array and (b) two-stage fine-grained systolic array.

with dynamic lengthy input sequences. Using the tiling concept [25], we construct a finer-grained dataflow allocation and systolic array scheduling scheme to reduce data skew (see blue dashed lines in Fig. 6(a)). The full attention reconstruction is performed by alternating between feature embedding loads and two-level dot-product for $M \times M$ tiles $G_{i,j}$ ($i = 1, 2, \dots, M; j = 1, 2, \dots, M$). As depicted in Fig. 6(b), the specific two-stage process can be described as follows:

- ① The forward-shifted feature embeddings are preferentially entered into the tiles $G_{i,i}$ distributed along the main diagonal.
- ② The output data from the bottom or right side of previous tiles is piped directly into the non-primary diagonal tiles $G_{i,j}$ ($i \neq j$) for the remaining outer-product accumulation computations.

For a 4×4 systolic array, the fine-grained scheme saves two cycles compared to the coarse-grained scheme. Furthermore, we can derive the latency consumed by the coarse-grained and fine-grained schemes for $N \times N$ attention reconstruction as T_c and T_f , when the feature embedding length is K .

$$T_c = 2N + K - 2$$

$$T_f = \left\lceil 2 \left\lfloor \frac{N}{M} \right\rfloor + K - 2, 2 \left\lceil \frac{N}{M} \right\rceil + K - 2 \right\rceil \quad (21)$$

where $\lfloor \cdot \rfloor$ and $\lceil \cdot \rceil$ denote the floor and ceil functions, respectively.

4.4 Fused Softmax Engine

To streamline the computation of weak-correlated attention, the fused softmax engine is engineered to enable efficient attention probability conversion for context aggregation. A non-linear operator, a mask generation module, and a mask-guided recomputation module work together in several steps to enable this conversion.

① **Non-linear Operator.** The 8-bit attention score, retrieved row-by-row from the OS-based SRAM-CIM, can be used to generate the exponential reduction vector $\exp(x - x_{\max})$. This process involves maximum search and reduction, followed by dequantization and exponential activation, where the exponential function is approximated by its fourth-order Taylor series.

② **Mask Generation.** The mask generation module conducts element-wise threshold comparison on the MSB of attention score vectors to generate binary mask vectors.

③ **Mask-guided Recomputation.** The masks determine if the exponential reduction value $\exp(x_i - x_{\max})$ for the i -th token should be fed into the accumulator and FIFO queue. Zero-mask values

prevent the current token from participating in accumulator calculations by setting its FIFO queue value to zero. After the mask-valid tokens accumulate $\sum_{i=1}^N \exp(x_i - x_{\max})$, the normalized results are recalculated using Equation 1. Finally, the 32-bit recomputed Softmax results are quantized again to yield the 8-bit probability.

5 Evaluation

5.1 Evaluation Methodology

We evaluate AESHA with benchmarks of two modalities using encoder-based transformer models. For BERT [1] and its sparse variants (BigBird [33], Sanger [14]), we test 8 language understanding tasks (MNLI, QNLI, QQP, SST2, STSB, RTE, COLA and MRPC) from the GLUE [34] datasets with the default sequence length of 128. For Vision Transformers (ViTs), we measure image recognition performance on 3 datasets (CIFAR10/100 and ImageNet1K [35]) where a CNN with the kernel and step sizes of 16 is used as a backbone to extract the features of 196 patches from 224×224 images.

We fine-tune pre-trained models from the Huggingface transformer library [36] when applying the pruning strategy to compress attention feature spaces with $\tau_{unit} = 0.1$. All tasks are executed using PyTorch v2.0 and CuDNN v11.8 on two Nvidia RTX 4090 GPUs. We adopt the fine-tuning parameters outlined in Sanger [14] for BERT and its variants. For ViTs, we employ the Adam optimizer with a learning rate of $3e-2$ and a batch size of 128. The fine-tuning epochs range from 5 to 15 depending on the downstream task.

We simulate the RRAM-CIM and SRAM-CIM macros using the peripheral setups under 32nm node technology in NeuroSim [37]. The read and write pulse times of the RRAM cells are taken from [38], whereas the on/off resistances of RRAM cells with the 4-bit cell precision [39] are extracted from [26]. We verify the SA-based LPU via Cadence Virtuoso and estimate the power of the OS-based attention reconstruction engine and other digital modules using Synopsys Design Compiler at 28nm technology. Additionally, we measure Nvidia GPU and Intel CPU's power by Nvidia-smi and PyRAPL, and latency using CUDA Event and Python timer.

5.2 Algorithm Performance

1) **Impact of Pruning Threshold.** We first analyze how pruning thresholds affect model performance. As depicted in Fig. 7(a), the model performance diminishes as the pruning threshold decreases. Reducing the pruning threshold causes eigenspace compression. Thus, the reconstructed attention matrix can not fully reflect the true token relevance distribution. BERT-Base and its sparse variants consistently exhibit performance deterioration across diverse tasks. Conversely, BERT-Large and ViTs demonstrate greater resilience to changes in the pruning threshold. Specifically, when the pruning threshold is set to 0.6, BERT-Base experiences performance declines ranging from 0.8% to 30.2% across different tasks. In contrast, BERT-Large's performance decreases ranging from 0.53% to 7.25%, and ViTs show a range of 1.14% to 9.55%.

Next, we analyze the impact of different pruning thresholds on the average compression ratio (ACR) of the weight parameters across three attention computation pipelines.

$$ACR_{\text{VANILLA}} = \frac{D}{KH}; ACR_{\text{CPSAA}} = \frac{D}{2K}; ACR_{\text{AESHA}} = \frac{D}{K} \quad (22)$$

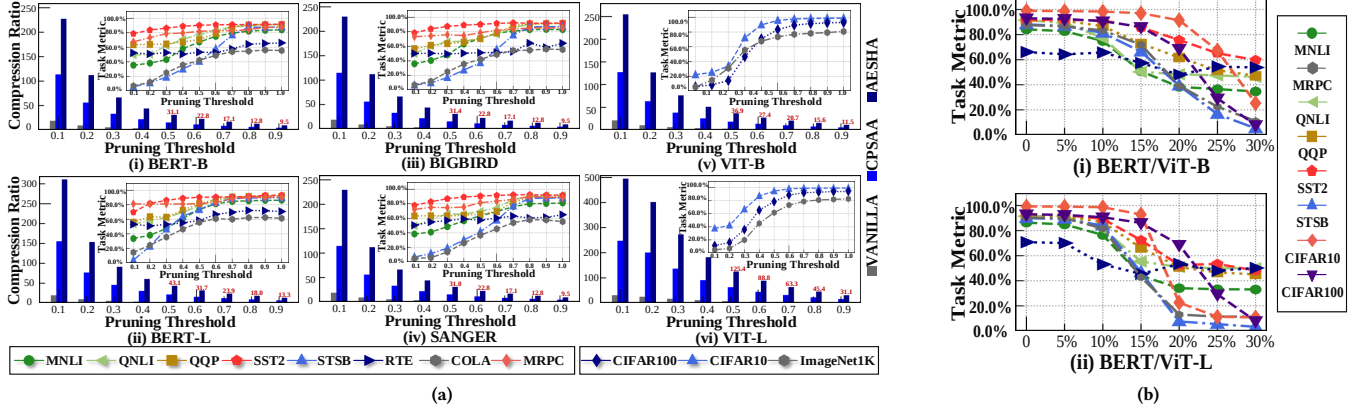


Figure 7: (a) Sensitivity of model performance and attention weight memory occupancy regarding pruning thresholds (without fine-tuning). (b) Sensitivity of model performance to RRAM variation.

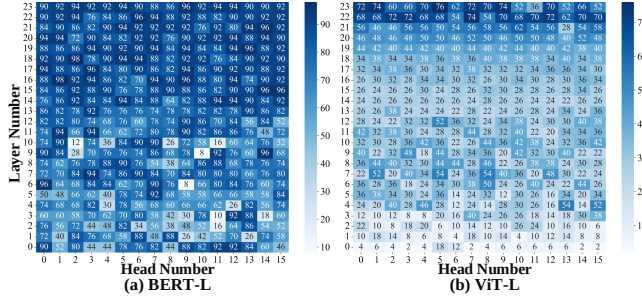


Figure 8: Visualization of Pruning Results ($\tau = 0.9$).

where D , K , and H represent the feature embedding dimension, the compressed dimension, and the number of heads respectively.

The ACR trends remain largely consistent for the same model across various tasks, while the compression potential of the eigenspace varies across different models. In conjunction with Fig. 7(a), By adopting the AESHA pipeline with the pruning threshold of 0.5, BERT-Base, BERT-Large, BiGBIRD, Sanger, ViT-Base, and ViT-Large achieve compression ratios of 31.1 $\times$, 43.1 $\times$, 31.4 $\times$, 31.0 $\times$, 36.9 $\times$, and 125.4 $\times$, respectively. These ratios are in comparison to the 2.59 $\times$, 2.69 $\times$, 2.62 $\times$, 2.58 $\times$, 3.08 $\times$, and 7.84 $\times$ compression achieved in the vanilla computation stack. Such compression results effectively reduce the memory footprint of static weights and the write demand of intermediate results. Subsequently, after fine-tuning the pruned models with the global optimization objective in Equation 17, the model's performance only degrades by 0.63% to 3.21% compared to the baseline models even at this aggressive pruning threshold.

2) Sensitivity Analysis. We visualize the distribution of multi-head attention eigenspace across different layers in the BERT-Large and ViT-Large at a pruning threshold of 0.9. As depicted in Fig. 8, the multi-head eigenspace of each layer in BERT-Large requires more eigenbases overall to satisfy the threshold constraint. In contrast, ViT-Large exhibits a sparser eigenspace distribution. Specifically, the multi-head eigenspace of shallow layers shows significant low-rank properties, while the deep layers require more eigenbasis to express the energy distribution of the attention weights. We attribute the low-rank phenomena to the intrinsic information representation differences between the two modalities. Images often

Table 2: Performance comparison of attention reconstruction engine on arrays of different granularity

Model / Metrics		32 $\times$ 32		64 $\times$ 64		128 $\times$ 128	
		EE	TP	EE	TP	EE	TP
ViT	Base	1.195	9.183	0.687	4.830	0.370	2.479
	Large	0.598	3.367	0.321	1.901	0.166	1.016
BERT	Base	3.054	8.922	1.792	5.140	0.977	2.782
	Large	2.966	8.664	1.730	4.968	0.941	2.680

Energy Efficiency (EE): TOPS/W, Throughput (TP): TOPS

have background noise, which contributes less to the overall energy distribution of the eigenspace, thus inducing low-rank features in shallow layers. In contrast, natural language encodes more dense information, resulting in a different low-rank behavior.

3) Impact of RRAM Variance. We evaluate the BERT and ViT model performance with varying degrees of Gaussian-distributed noise by introducing noise to fixed weights within the RRAM-CIM feature transformation engine. Fig. 7(b) shows that the model performance remains stable at 5% device variation. However, higher variations lead to more drastic performance degradation, especially noticeable in BERT/ViT-Large where the error caused by device non-ideality can be propagated and accumulated layer by layer.

5.3 Hardware Performance

1) Impact of Array Granularity. We measure the energy efficiency and throughput of the attention reconstruction engine across three different granularity levels of a systolic array. As depicted in Tab. 2, we observe that the attention reconstruction achieves higher energy efficiency: ViT can achieve 3.49 $\times$ energy efficiency enhancement and 3.70 $\times$ throughput improvement on the 32 $\times$ 32-grained systolic array, respectively, while BERT can achieve 3.15 $\times$ energy efficiency enhancement and 3.23 $\times$ throughput improvement. Two reasons account for this performance improvement. First, the power consumption induced by the dataflow scheduling control in the fine-grained scheme contributes much less to the total energy than the delay impact induced by data skew in the coarse-grained array. Second, the delay caused by data skew can be effectively mitigated in the fine-grained scheme by forwarding cached back-level data and enabling multiple tiles to work in parallel.

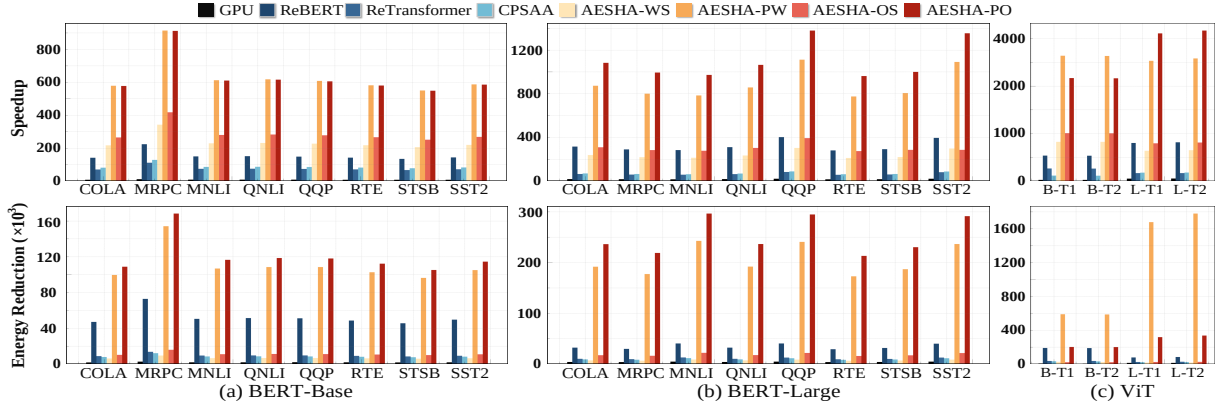


Figure 9: Speedup and energy reduction of self-attention layer across different pipeline-optimized work concerning CPU baseline. AESHA-WS employs conventional WS-based SRAM-CIM for attention computation, involving the same CWC operations as ReTransformer. AESHA-OS adopts OS-based SRAM-CIM for attention reconstruction. Additionally, AESHA-PW and AESHA-PO implement pruning strategies. B/L denote ViT-Base/Large models, while T1/T2 represent CIFAR-10/100 dataset.

Table 3: Comparison of throughput and energy efficiency against state-of-the-arts

Models&Pipelines	ReBERT [24]	ReTransformer [15]	CPSAA [16]	AESHA-PW	AESHA-PO
Throughput (GOPS)					
ViT-B	134.19	107.77	121.86	97.63	176.12
ViT-L	128.57	79.42	44.28	136.82	149.07
BERT-B	134.93	108.16	121.92	97.69	176.47
BERT-L	128.95	78.90	44.64	70.58	149.22
Energy Efficiency (TOPS/W)					
ViT-B	0.96	0.24	0.22	2.65	3.57
ViT-L	0.96	0.19	0.18	2.59	4.09
BERT-B	0.90	0.23	0.22	1.44	2.28
BERT-L	0.93	0.17	0.18	1.50	2.96

2) Impact of Pipeline Designs. We evaluate the speedup and energy efficiency of four AESHA-based pipeline-optimized architectures (i.e., AESHA-WS, AESHA-OS, AESHA-PW and AESHA-PO) and prior arts over the CPU baseline for self-attention processing of BERTs and ViTs. The detailed definition of these AESHA variants is given in the caption of Fig. 9. For optimized pipelines without adopting the pruning strategy, AESHA-WS and AESHA-OS can achieve higher speedups than ReBERT on BERT/ViT-Base models. In contrast, speedups are comparable in BERT/ViT-Large counterparts. Compared to the general purpose computing platforms CPU, GPU and CIM-based ReBERT, ReTransformer and CPSAA, AESHA-PO achieves maximum inference speedups of 3170.0 $\times$, 95.6 $\times$, 4.1 $\times$, 19.0 $\times$, 19.2 $\times$ on a single attention layer. In the BERT/ViT-Large models, AESHA-PO exhibits stronger speedup gains than AESHA-PW. Two main reasons contribute to this phenomenon. First, adopting the energy-distribution-based pruning strategy on deeper models can achieve higher compression ratios and parameter reduction. Second, AESHA-PO consumes fewer data movements and computation cycles than the compute-write-compute manner in AESHA-PW, particularly in cases of low feature embeddings.

AESHA-PW and AESHA-PO both save more energy across models and benchmark datasets due to the pruned operations. Compared to CPU, GPU, ReBERT, ReTransformer and CPSAA, AESHA-PO reduces energy consumption on the attention layers by 336.0K $\times$, 12.6K $\times$, 7.3 $\times$, 23.1 $\times$, 25.8 $\times$. AESHA-PO achieves energy reductions of 8.74% $\sim$ 9.36% more than AESHA-PW on BERT-Base and 22.18% $\sim$ 23.45% more reduction on BERT-Large. However, AESHA-PO exhibits less energy-saving performance on ViTs compared to AESHA-PW, since the OS-based dynamic attention reconstruction

engine consumes quadratically more energy with token numbers. Notably, ViTs require longer input sequences than BERTs.

3) End-to-end Comparison. For a fair comparison, we evaluate the throughput and energy efficiency of Transformers across different pipeline optimization approaches with the same hardware configurations. As shown in Tab. 3, AESHA-PO yields the best overall throughput and energy efficiency. Compared to state-of-the-art methods, AESHA-PO achieves 176 GOPS and 149 GOPS throughput on ViT/BERT-Base and ViT/BERT-Large models, indicating a throughput improvement of 16.33% $\sim$ 31.24% from ReBERT. Additionally, AESHA-PO demonstrates 3.72 $\times$, 21.53 $\times$, 22.72 $\times$ and 1.97 $\times$ energy efficiency improvements compared to ReBERT, ReTransformer, CPSAA and AESHA-PW. The energy efficiency values can be further enhanced if multi-bit input is allowed for RRAM-CIM.

6 Conclusion

In this work, we present AESHA, an RRAM and SRAM-based algorithm and architecture co-design framework to accelerate the self-attention-based Transformer models efficiently. On the algorithm level, we propose an eigen-decomposition-based attention computation pipeline and an energy-distribution-based structured pruning strategy in order to enhance attention computation efficiency while minimizing the memory footprint and RRAM write counts. On the architecture level, we introduce a heterogeneous CIM implementation featuring hybrid dataflow to balance the workload of static feature transformation and dynamic attention reconstruction. Furthermore, we design a fine-grained systolic array to improve the throughput and energy efficiency. Our AESHA achieves a compression ratio of attention weight parameters of 2.5 $\times$ $\sim$ 2.69 $\times$ on BERTs and 3.07 $\times$ $\sim$ 7.84 $\times$ on ViTs with minimal performance degradation. In addition, our AESHA achieves superior latency and power efficiency compared to prior arts, as evidenced by achieving 31.24% higher throughput and 3.72 $\times$ better energy efficiency than ReBERT. AESHA presents a promising solution for lightweight, low-power, high-throughput edge transformer inference applications.

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